

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

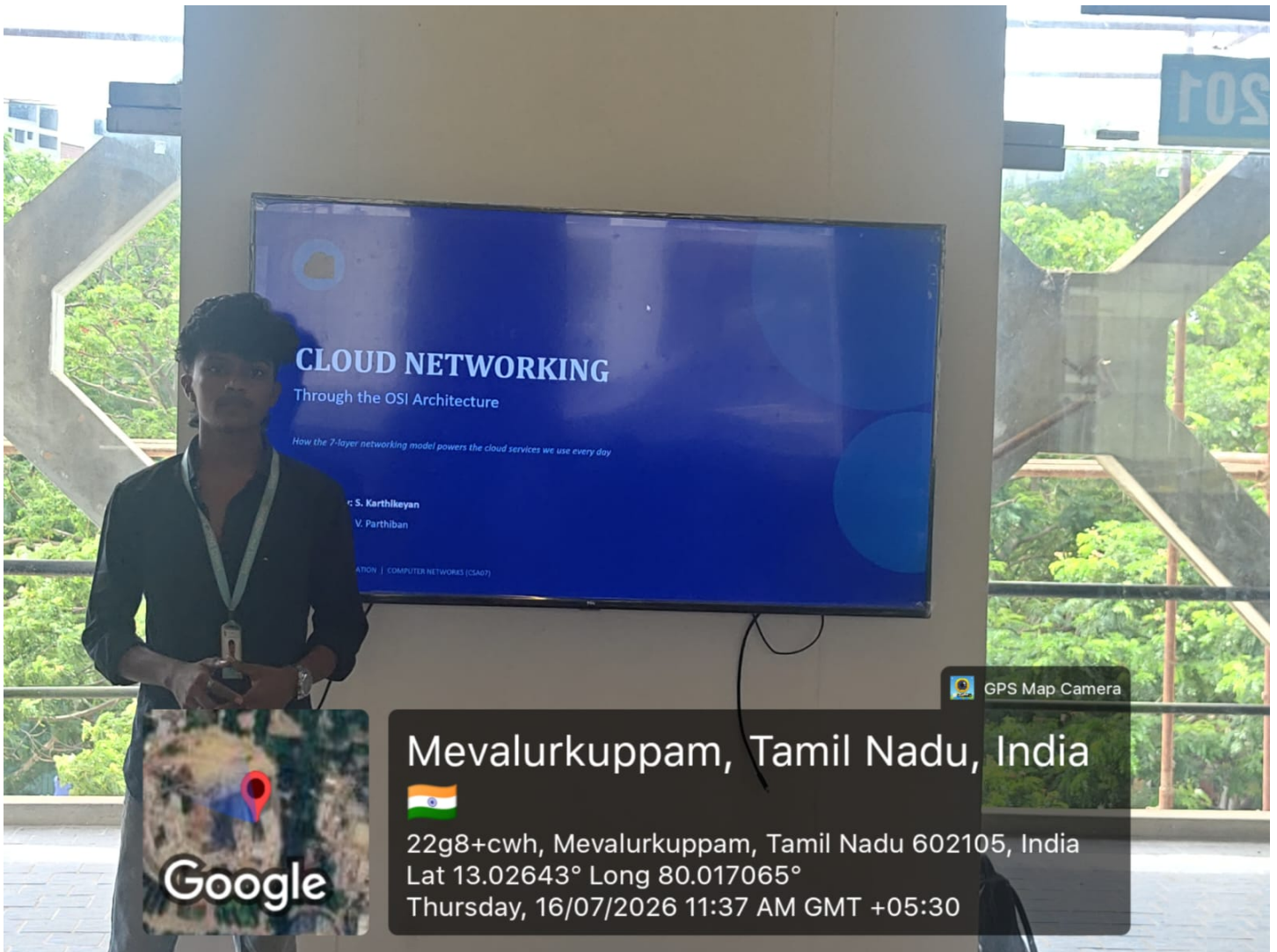
Code of Conduct:

I S. Karthikeyan / 192419048 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A



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CLOUD NETWORKING

Through the OSI Architecture

How the 7-layer networking model powers the cloud services we use every day

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Guided by: Dr. V. Parthiban

What Is Cloud Networking?

Cloud networking delivers computing resources — servers, storage, applications — over the internet instead of local hardware. Every request a user makes still has to travel through the same reliable, layered process defined by the OSI Model.



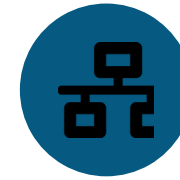
Global Access

Users connect to cloud services from anywhere via the internet.



Shared Resources

Servers, storage and apps run in remote data centers, not on-site.



Layered Delivery

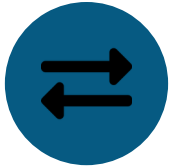
Every connection still passes through the 7 OSI layers.

The OSI Model: 7 Layers

Data travels down these layers to be sent, and up them again to be received.

7	Application	User-facing apps: browsers, cloud dashboards
6	Presentation	Data formatting, encryption (TLS/SSL)
5	Session	Opens & maintains connections to cloud servers
4	Transport	Reliable delivery via TCP, load-balanced streams
3	Network	IP addressing & routing across data centers
2	Data Link	Framing within a data center's local network
1	Physical	Cables, fiber & signals inside server racks

How the Cloud Puts OSI to Work



Application Layer

APIs & web apps (HTTP/HTTPS) let you access cloud storage, email and dashboards.



Presentation Layer

TLS encryption protects data as it moves to and from the cloud provider.



Network Layer

IP routing directs traffic to the nearest data center for faster response.



Transport Layer

TCP ensures files and requests arrive complete and in the correct order.

Benefits & Challenges



Benefits

- ✓ Scalable — add capacity without new hardware
- ✓ Reliable — OSI layers ensure error-checked delivery
- ✓ Cost-effective — pay only for what you use
- ✓ Fast — routing sends data via the shortest path



Challenges

- ✗ Latency — extra network hops can slow responses
- ✗ Security — more layers mean more points to protect
- ✗ Downtime risk — depends on internet connectivity
- ✗ Complex troubleshooting across many layers



Key Takeaway

The cloud feels invisible and instant, but every click still relies on the same 7 OSI layers working together — from physical cables to the application on your screen.

- OSI gives cloud networks a common, reliable structure
- Each layer solves one problem — together they deliver speed & trust
- Understanding OSI helps troubleshoot real cloud issues

Thank You